

Abstract

The aim of this sub-project is to develop an image processing component to support low-cost automated greenhouses. The image processing application must enable the greenhouse system to establish various aspects about the plants from the images taken. These aspects include plant health, plant growth and plant disease identification from the images of different parts of the plants in the greenhouse.

Fill out the introduction.

What is the purpose of this section? To introduce background information about phenotyping in relation to plants. - Why it is useful? - Relations to current study at hand?

In food security, and plant breeding it is desirable to identify cultivars which have desirable resistances against abiotic and biotic stresses. For example, given two cultivars A and B in a water scarce environment we would like to identify the which plant fares the best. Best in the sense of biomass, resistant to disease, metabolic health. All of these observable properties of the cultivars that can be quantified and measure enable us to make decisions to choose and breed cultivars with genes most suited for specific growing environments whilst minimizing loss and maximizing yield. This process or field of study which provides the mechanism for this is called Plant Phenotyping.

What is plant phenotyping? It is of vital importance in agriculture to identify the best cultivars for certain environments, especially in the context of food security where adverse abiotic and biotic stresses affect crop yield.

TODO: Find a reference for this For example two cultivars A, and B in some water scarce environment could have vastly different resistances to stress and therefore potentially different crop yields. Obviously as a plant breeder you would like to choose genes from the cultivar which gives you the most crop yield, and breed future cultivars to have this desirable trait. Crop yield is a simple case, but what if we would like to optimize resistance to disease? Or resistance to water scarcity? We need a quantitative metrics that allows us to measure observable traits that predict properties/outcomes of the plants we care about. The field of study and mechanism for this is called plant phenotyping.

The concept of plant phenotyping or term was introduced by Johannsen and is loosely defined as Definition 1. However over the years as we developed more advanced chemical and analytical ways to measure the metabolic and chemical processes of plants this definition in the literature has expanded in scope. All the other definitions

Why plant phenotyping in the context of this study? Fill this out

My definition: Plant phenotyping falls under the larger field of plant phenomics. A phenotype is loosely a type of a phenomena specific organism may exhibit. Two different organisms may exhibit a phenotype in two drastically different ways depending on the interaction between the phenotype, environment, and the organisms genes. Phenotypical properties of organisms are multifaceted, for example plant metabolic health is a phenotype depending on the environment into terms of water, and heat which is abiotically. Additionally it might be dependent on the plants genes. Some plants might be more metabolically healthy in more humid conditions.

Definition 1: “All ‘types’ of organisms, distinguishable by direct inspection or only by finer methods of measuring or description, may be characterized as ‘phenotypes’. [3]

Definition 2: The phenotype can be seen as the combination of all the morphological, physiological, anatomical, chemical, developmental, and behavioral characteristics that, when put together, represent the individual organism. Phenotyping, which is the process of characterizing the phenotype [2]

Definition 3: For the purposes of this review, we refer to phenotyping as the set of methodologies and protocols used to measure plant growth, architecture, and composition with a certain accuracy and precision at different scales of organization, from organs to canopies. The goal of modern plant phenotyping is to deliver quantitative data on the dynamic responses of plants to the environment. [1]

Definition 4: The broad evaluation of complex plant traits like growth, development, tolerance, resistance, architecture, physiology, ecology, and yield as well as the fundamental measurement of individual quantitative parameters that serve as the foundation for more complex traits is known as plant phenotyping [5]

Definition 5: Plant phenotyping is the comprehensive assessment of complex plant traits such as growth, development, tolerance, resistance, architecture, physiology, ecology, yield, and the basic measurement of individual quantitative parameters that form the basis for more complex traits [4]

Fill out the Image processing techniques section.

Fill out the applications in low cost greenhousing techniques secton.

Fill out the conclusion section.

References

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